

$^4\text{He}(^{30}\text{S},\alpha)$:resonances **2018Ka11**

2018Ka11: 48.4-MeV $^{20}_{10}\text{S}$ was produced via the $^3\text{He}(^{28}\text{Si}, ^{30}\text{S})\text{n}$ reaction, separated by the CRIB separator at RIKEN, passed through two PPACs, and entered an active target filled with 90% $^{\text{nat}}\text{He}$ + 10% CO_2 gas at 194.2 Torr 5. Scattered α particles were detected using the active target and 500- μm thick SSDs. Measured E_α , I_α , and $^{30}\text{S}+\alpha$ excitation functions at $E_{\text{c.m.}}=2.0\text{--}5.5$ MeV. Deduced 3 resonances, J , π , L_α , Γ_α , and Γ_p from multichannel R-Matrix analysis.

 ^{34}Ar Levels

<u>E(level)</u>	<u>J^π</u>	<u>L</u>	<u>Comments</u>
11092 85	($2^+, 4^+$)	2,4	$E_{\text{c.m.}}=4353$ keV. For $J^\pi=2^+$, $\Gamma_\alpha=20$ keV $+80-17$, $\Gamma_p=25$ keV $+500-20$. $\omega\gamma(\alpha,p)=100$ keV. For $J^\pi=4^+$, $\Gamma_\alpha=0.5$ keV $+14-4$, $\Gamma_p=0.3$ keV $+35-3$.
11518 89	2^+	2	$E_{\text{c.m.}}=4779$ keV. $\Gamma_\alpha=100$ keV $+120-60$, $\Gamma_p=210$ keV $+600-170$. $\omega\gamma(\alpha,p)=500$ keV.
12079 95	(2^+)	2	$E_{\text{c.m.}}=5340$ keV. $\Gamma_\alpha=260$ keV $+400-120$, $\Gamma_p=340$ keV $+550-200$. $\omega\gamma(\alpha,p)=1300$ keV.